

Quarry: Technical Architecture Specification

Local Semantic Search for AI Agents

July 2026

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1 Overview

Quarry is a local semantic search engine. It ingests documents in 20+ formats, embeds them with a local ONNX model, stores vectors in LanceDB, and serves semantic search to AI agents (Claude Code, Claude Desktop) and a macOS menu bar app.

The system is **daemon-first** (DES-031 v2.2, ACCEPTED, in implementation): a single **quarryd** process holds the engine, and the CLI, MCP, and library are clients that reach it over a versioned wire protocol — the daemon is the intermediary service layer. As of v2-3a the engine runs only in **quarryd** (the **quarry serve** subcommand is retired), loopback requests are authenticated with a **mode-0600 serve.token**, and the CLI remote paths resolve that token through **ClientConfig**. The full client tier (**QuarryClient**) and **quarry mcp-as-client** are in flight (v2-3/v2-4); until they land, the local-mode CLI and **quarry mcp** still load the engine in-process.

1.1 Design Principles

- P1. Daemon-first.** The **quarryd** process is the single engine (embedding model, LanceDB, pipeline, sync registry); the CLI, MCP, and library are clients over a versioned REST contract (DES-031 v2.2, ACCEPTED, in implementation). The prior *library-first* principle — the core library did all work and the surfaces were thin wrappers — described the as-built state DES-031 v2 replaces.
- P2. Local by default.** Everything runs on the user’s machine. No API keys, no cloud accounts. The embedding model downloads once (~120 MB, int8 quantized).
- P3. One daemon, many clients.** A single **quarryd** process loads the embedding model once and serves all sessions over the REST wire; supervised by **launchd KeepAlive** / **systemd Restart=always**.
- P4. Fire-and-forget I/O.** Side-effect MCP tools return immediately and process in a bounded thread pool. Search is synchronous.
- P5. Named databases.** Each database is fully isolated: its own LanceDB directory, sync registry, and vector index. Work/personal separation without configuration files.

2 System Architecture

2.1 Deployment Topology

Single machine (localhost TLS):

```
        stdio wss:// (TLS)
Claude Code <-----> mcp-proxy <-----> quarryd
        MCP JSON-RPC (~5 MB Go) pinned CA cert (one daemon)
```

Remote server (GPU):

```
Mac (client) Linux (server)
        stdio wss:// (TLS)
Claude Code <-----> mcp-proxy <-----> quarryd
        MCP JSON-RPC (~5 MB Go) pinned CA cert (GPU daemon)

CLI -----> HTTPS GET/POST -----> quarryd
        Bearer + pinned CA
```

Without the proxy, every Claude Code tab spawns a separate Python process, each loading the embedding model into ~200MB of RAM. With it, startup is instant and state is shared across all sessions. Installed services use TLS with TOFU certificate pinning (DES-020) — even on localhost.

When a remote server is configured (DES-021), the CLI routes all data commands over HTTPS to the server, with only authentication and administration commands remaining local. **find**, **show**, **status**, **ingest**, **remember**, **delete**, **sync**, all **list** subcommands, and directory **register/deregister** route remotely. **use** is local by design: it sets the client’s persistent default database, but the remote server is fixed to the database it was started with, so switching databases is a client-side setting rather than a remote call.

2.2 Surface Summary

Surface	Transport	Purpose
CLI	Direct library calls	Interactive use from terminal
MCP server	stdio or WebSocket	Claude Code/Desktop tools
HTTP server	REST JSON + WS /mcp	Menu bar app + mcp-proxy
Plugin	Hook shell commands	Auto-register, auto-ingest, compaction safety

3 Module Responsibilities

3.1 Data Layer (src/quarry/db/)

The database facade composes five subsystems behind a single entry point (**Database**), so callers pass one object rather than a raw LanceDB connection.

Module	File	Responsibility
Facade	src/quarry/db/facade.py	Database — single entry point composing the store, search, catalog, schema, and optimizer subsystems (DES-031).
Chunk store	src/quarry/db/chunk_store.py	Insert, progressive-flush, delete, and count operations on the chunks table.
Chunk search	src/quarry/db/chunk_search.py	Vector-similarity search under the cosine metric.
Chunk catalog	src/quarry/db/chunk_catalog.py	List documents, list collections, and fetch page text.
Optimizer	src/quarry/db/optimizer.py	Table compaction, FTS index rebuild, and per-collection indexing.
Schema	src/quarry/db/schema.py	Chunks-table schema and idempotent migration (agent-memory columns, FTS index).
Storage	src/quarry/db/storage.py	LanceDB connection, database discovery, and size formatting.

Supporting types: `src/quarry/models.py` (immutable `PageContent`, `Chunk`, `PageAnalysis` data-classes), `src/quarry/types.py` (structural Protocols for LanceDB, OCR, and embedding backends), `src/quarry/results.py` (`SearchResult`, `SearchFilter`, `WORST_CASE_DISTANCE`), and `src/quarry/config.py` (`Settings`, `resolve_db_paths()`, ONNX model constants).

3.2 Ingestion Pipeline (src/quarry/ingestion/)

Module	File	Responsibility
Pipeline	src/quarry/ingestion/pipeline.py	Orchestrator: dispatch by format, chunk, embed, store. Entry points <code>ingest_document</code> , <code>ingest_content</code> , <code>ingest_url</code> .
Chunker	src/quarry/ingestion/chunker.py	Split page text into overlapping chunks; respects page and heading boundaries.
URL ingester	src/quarry/ingestion/url_ingester.py	Fetch, parse HTML, chunk, embed, store. <code>ingest_url</code> takes an opt-in <code>content_scrubber</code> (see PII redaction).
Web fetch	src/quarry/ingestion/web_fetch.py	Fetch a URL over HTTP(S) and return validated HTML text.
Ingest stats	src/quarry/ingestion/ingest_stats.py	Per-format ingest counters merged into an ingest result.

Format extraction lives in `src/quarry/extractors/`; embedding and OCR provider selection in `src/quarry/ingestion/provider.py`, `src/quarry/ingestion/backends.py`, and `src/quarry/ingestion/ocr_local`.

3.3 Retrieval Seam (src/quarry/retrieval/)

Hybrid search runs through one parameterizable seam shared by every surface and by the eval harness, so the production path and the committed baseline cannot drift.

Module	File	Responsibility
Service	src/quarry/retrieval/service.py	SearchService — the single call every surface (CLI, HTTP, MCP) makes for search. Kills remote/local divergence (bug class 3).
Hybrid	src/quarry/retrieval/hybrid.py	HybridRetriever — vector + BM25 channels fused with RRF, behind a config.
Config	src/quarry/retrieval/config.py	RetrievalConfig — frozen retrieval knobs (RRF k , over-fetch, cosine metric, decay rate, reranker). Defaults reproduce shipped search bit-for-bit.
Protocols	src/quarry/retrieval/protocols.py	Retriever and Reranker structural interfaces (one method each).
Fusion	src/quarry/retrieval/fusion.py	RrfFusion — Reciprocal Rank Fusion with optional temporal decay.
Reranker	src/quarry/retrieval/reranker.py	NullReranker — pass-through Null Object when reranking is off.

3.4 Format Processors (src/quarry/extractors/)

Each extractor implements a common Protocol (src/quarry/extractors/protocol.py) and converts a source format into `list[PageContent]`:

Module	File	Formats
PDF	src/quarry/extractors/pdf_extractor.py	PDF (text via PyMuPDF, OCR fallback for image pages)
Image	src/quarry/extractors/image_extractor.py	PNG, JPG, TIFF, BMP, WebP (OCR)
Text	src/quarry/extractors/text_extractor.py	TXT, MD, LaTeX, DOCX
Code	src/quarry/extractors/code_extractor.py	30+ languages (tree-sitter section splitting)
HTML	src/quarry/extractors/html_extractor.py	HTML (boilerplate stripping, Markdown conversion)
Spreadsheet	src/quarry/extractors/spreadsheet_extractor.py	XLSX, CSV (LaTeX tabular serialization)
Presentation	src/quarry/extractors/presentation_extractor.py	PPTX (slide-per-chunk with notes)

3.5 Backends

Module	File	Responsibility
Factory	src/quarry/ingestion/backends.py	Thread-safe factory with double-checked locking. <code>get_ocr_backend()</code> and <code>get_embedding_backend()</code> return cached singletons.
OCR	src/quarry/ingestion/ocr_local.py	Local OCR via RapidOCR (offline, no setup).
Embeddings	src/quarry/embeddings.py	Local ONNX embedding (snowflake-arctic-embed-m-v1.5, 768-dim, 512 tokens). Vectors are L2-normalized to unit length.
Provider	src/quarry/ingestion/provider.py	ONNX Runtime execution provider auto-detection (CUDA/FP16 or CPU/int8).

3.6 Capture and Redaction

Session transcripts and WebFetch auto-captures are scrubbed at write time before any bytes reach a git-tracked (and soon pushed) surface (DES-036).

Module	File	Responsibility
Capture writer	src/quarry/capture.py	CaptureWriter — the single choke point both .md producers route through. Scrubs, then atomically writes.
Scrubber	src/quarry/scrub.py	Scrubber — ordered redaction passes (secrets, paths, emails, hostname, profanity). Idempotent.
Scrub rules	src/quarry/scrub_rules.py	Secret-detection regex catalog applied by the scrubber.
Capture URL	src/quarry/capture_url.py	CaptureUrl — strips user-info/query/fragment from a captured URL's metadata form.
Web capture	src/quarry/web_capture.py	Parse a PostToolUse WebFetch payload into URL + fetched content.
Transcript	src/quarry/transcript.py	The session transcript file and its derived identifiers.
Backfill	src/quarry/backfill.py	Backfill historical session transcripts through the capture writer.

3.7 Fetch Safety (SSRF)

Every outbound fetch — the leaf-page fetch and the sitemap crawl — runs the address-safety gate (**UrlSafetyCheck**) against the *resolved* address on *every* hop, fail-closed, before the socket opens (DES-043). The route boundary validates only the initial caller-supplied URL; the gap this closes is that redirect targets and sitemap-discovered URLs never reached that gate — an authenticated caller could otherwise reach 169.254.169.254, loopback, or a private range via a redirect from a public server or an internal URL listed in a sitemap.

Module	File	Responsibility
URL safety	src/quarry/url_safety.py	UrlSafetyCheck — classifies a URL's resolved address (rejects link-local, loopback, RFC-1918, CGNAT, unspecified, IPv4-mapped-IPv6, and the NAT64 well-known prefix; checks all <code>getaddrinfo</code> records). In core, so the daemon route and the fetch layer share one classifier.
Redirect guard	src/quarry/ingestion/ssrf_redirect.py	SsrfGuardedRedirectHandler plus the shared <code>GUARDED_OPENER</code> : re-gates every 30x <code>Location</code> before it is opened.
Sitemap client	src/quarry/sitemap_web_client.py	GatedSitemapWebClient — USP's AbstractWebClient , gating every sitemap-index, <code>robots.txt</code> , and sub-sitemap fetch at every recursion depth (a block is non-retryable, so USP never connects).

A residual DNS-rebind TOCTOU (the gate resolves, the socket re-resolves) is the tracked validate-then-pin follow-up (quarry-kmzo/quarry-ljym).

3.8 Sync and Registry

Module	File	Responsibility
Sync	<code>src/quarry/sync.py</code>	Directory sync: discover files, compute delta (new/changed/deleted), ingest/delete accordingly.
Discovery	<code>src/quarry/sync_discovery.py</code>	Walk, filter, and hash files for the sync delta.
Ingest	<code>src/quarry/sync_ingest.py</code>	Progressive per-collection ingest (producer/consumer with within-file resume).
Registry	<code>src/quarry/sync_registry.py</code>	SQLite registry of watched directories and per-file state.

3.9 Surfaces

Module	File	Responsibility
CLI	<code>src/quarry/_main_.py</code>	Typer CLI with rich formatting. Synchronous. All commands: resolve settings → call core → format output.
MCP Server	<code>src/quarry/mcp_server.py</code>	FastMCP server. Read-only tools are synchronous. Side-effect tools use fire-and-forget.
HTTP Server	<code>src/quarry/http_server.py</code>	Starlette ASGI app on uvicorn. REST endpoints + WebSocket /mcp.
Hooks	<code>src/quarry/hooks.py</code>	Claude Code plugin hooks: session-start (auto-register), post-web-fetch (auto-ingest), pre-compact (capture transcript). All fail-open.
Hook Entry	<code>src/quarry/_hook_entry.py</code>	Lightweight hook dispatcher that bypasses full CLI import chain. Dispatches via <code>sys.argv</code> with lazy imports (~0.1s vs ~1.5s for full CLI).
Remote	<code>src/quarry/remote.py</code>	Remote server config: mcp-proxy config read/write, TOFU CA cert fetch and storage, connection validation.
Remote client	<code>src/quarry/remote_client.py</code>	<code>RemoteClient</code> — authenticated HTTP client the CLI uses to route data commands to a remote daemon over pinned-CA HTTPS.
Daemon	<code>src/quarry/service.py</code>	quarryd lifecycle via <code>launchd/systemd</code> .

3.10 Support

Module	File	Responsibility
Formatting	<code>src/quarry/formatting.py</code>	Pre-formatted plain text output using unicode box-drawing. Used by both CLI and MCP for consistent output.
Sitemap	<code>src/quarry/sitemap.py</code>	Sitemap discovery and URL extraction for web ingestion.
Doctor	<code>src/quarry/doctor.py</code>	Health checks and install flow: model availability, database access, MCP configuration, ethos extension setup.
Enable	<code>src/quarry/enable.py</code>	Project activation and deactivation: register directory, bootstrap ethos memory extensions, write per-project config.
TLS	<code>src/quarry/tls.py</code>	TLS certificate generation: self-signed EC P-256 CA and server cert with atomic file writes and TOFU-safe CA reuse.
Proxy	<code>src/quarry/proxy.py</code>	<code>mcp-proxy</code> binary download, SHA256 verification, and platform-specific installation.
Logging	<code>src/quarry/logging_config.py</code>	Structured logging configuration: rotating file handler + stderr, <code>QUARRY_LOG_LEVEL</code> override.

4 Daemon Model

4.1 Why a Daemon

The embedding model (snowflake-arctic-embed-m-v1.5, int8 quantized ONNX) occupies ~200 MB of RAM and takes ~250 ms to load. MCP's stdio transport model spawns a new process per session. Without a daemon, five Claude Code tabs means five copies of the model in memory and five cold starts.

`quarryd` loads the model once and keeps it resident, respawned by `launchd KeepAlive` / `systemd Restart=always`.

4.2 Process Boundary and Loopback Authentication (v2.2)

`quarryd` (`src/quarry/daemon/launcher.py`) is a dedicated entry point — the sole engine process, making the client/engine boundary a hard *process* split. Only `src/quarry/daemon/` imports the engine; the `quarry serve` subcommand is deleted.

At startup `quarryd` refuses a remote-reachable bind that carries no operator key (an auto-generated token would be false security — it is unreadable by the remote clients that need it), and for a loopback bind mints a 256-bit `serve.token` when none is supplied. The daemon writes that token mode-0600 beside `serve.port` (shared `src/quarry/run_dir.py`) via an atomic `os.open(0o600) + tmp-rename`, and requires it on every request — closing the multi-user-host exposure where any local UID could reach the unauthenticated daemon on 127.0.0.1. Loopback classification (`src/quarry/net.py`, `LoopbackPolicy`) is `ipaddress`-based: `localhost/::1/127.0.0.0/8` are loopback, `0.0.0.0` and unresolved names fail closed.

Clients resolve the target through `ClientConfig` (`src/quarry/client/config.py`): for a loopback URL the bearer is read *live* from `serve.token` (it rotates every daemon restart, so a stored token is never trusted); for a remote URL the stored bearer is kept. A missing loopback token fails closed with an actionable error rather than a bare 401. `ClientConfig` is the client tier's first piece; the full `QuarryClient` and `quarry mcp-as-client` land in v2-3/v2-4.

The daemon serves the **REST API only** under `/v1` (the in-daemon MCP route was removed in PR-1); its request-admission trust model combines the `JsonContentTypeGuard` — an ASGI CSRF choke point that rejects any POST without `Content-Type: application/json` (a browser cannot send that cross-origin without a preflight the CORS policy refuses) — with the `serve.token` bearer above. The guard blocks a forged cross-origin write; the token blocks an unauthorized local reader.

4.3 mcp-proxy Architecture

`quarry install` downloads `mcp-proxy` (SHA256-verified, platform-specific binary) and configures MCP clients to use it. The proxy:

1. Reads MCP JSON-RPC from stdin.
2. Forwards to `wss://localhost:8420/mcp` over WebSocket (TLS, pinned CA — even on localhost).
3. Relays responses back to stdout.

Each WebSocket connection gets its own `asyncio Task` with a `ContextVar` for database selection (`_db_name`), so `use("work")` in one session does not affect others.

4.4 CLI and the Daemon

Under DES-031 v2 (ACCEPTED, in implementation) the CLI is a thin client: it builds a request, calls the daemon over the wire protocol via `QuarryClient`, and renders the response — it does not load the embedding model or open LanceDB in-process. The prior design had `quarry find "margins"` load the model in-process and query LanceDB directly; that local-engine path is removed as the data commands migrate to `QuarryClient`.

5 MCP Tool Surface

Tool	Key Parameters	Purpose	Exec
<code>find</code>	<code>query</code> , <code>limit</code> , <code>collection</code>	Semantic search	Sync
<code>ingest</code>	<code>source</code> , <code>overwrite</code> , <code>collection</code>	Ingest file or URL	Bg
<code>remember</code>	<code>content</code> , <code>document_name</code>	Ingest inline text	Bg
<code>list</code>	<code>kind</code>	List indexed data	Sync
<code>show</code>	<code>document_name</code> , <code>page_number</code>	Document metadata or page	Sync
<code>delete</code>	<code>name</code> , <code>kind</code>	Delete document or collection	Bg
<code>register_directory</code>	<code>directory</code> , <code>collection</code>	Register for sync	Bg
<code>deregister_directory</code>	<code>collection</code> , <code>keep_data</code>	Remove registration	Bg
<code>sync_all</code>	—	Sync all registered dirs	Bg
<code>status</code>	—	Database stats	Sync
<code>use</code>	<code>name</code>	Switch active database	Sync

Bg = background (`ThreadPoolExecutor`).

Background tools use a bounded `ThreadPoolExecutor(max_workers=4)`. They return an optimistic response immediately. Settings and database connections are snapshotted at the tool boundary and passed into background functions — this avoids a race condition where the mutable `_db_name` `ContextVar` could change between the tool call and the background thread's execution. Exceptions in background threads are logged via `logger.exception()`, not raised.

6 HTTP Server

6.1 ASGI Application

Starlette app built by `build_app()` factory, served by `uvicorn`. REST handlers are sync functions — Starlette auto-detects this and runs them in its threadpool. The `/mcp` WebSocket endpoint is async.

Default port: 8420 (`DEFAULT_PORT` in `src/quarry/config.py`), overridable with `--port`.

6.2 Endpoints

Path	Method	Purpose	Auth	Model
/health	GET	Server uptime and status	No	sync
/ca.crt	GET	CA certificate PEM (TOFU)	No	sync
/search	GET	Hybrid search	Yes	sync
/show	GET	Document metadata or page text	Yes	sync
/documents	GET	List documents	Yes	sync
/documents	DELETE	Delete document (returns 202)	Yes	async
/collections	GET	List collections	Yes	sync
/collections	DELETE	Delete collection (returns 202)	Yes	async
/registrations	GET	List directory registrations	Yes	sync
/registrations	POST	Register directory (returns 202)	Yes	async
/registrations	DELETE	Deregister directory (returns 202)	Yes	async
/databases	GET	List named databases	Yes	sync
/status	GET	Database statistics	Yes	sync
/use	POST	Switch active database	Yes	sync
/remember	POST	Ingest inline content (returns 202)	Yes	async
/ingest	POST	Ingest URL (returns 202)	Yes	async
/sync	POST	Sync registrations (returns 202)	Yes	async
/tasks/{task_id}	GET	Poll task status (unified)	Yes	sync
/sync/{task_id}	GET	Alias for /tasks/{task_id}	Yes	sync
/ingest/{task_id}	GET	Alias for /tasks/{task_id}	Yes	sync
/mcp	WS	MCP JSON-RPC over WebSocket	Yes	—

See Appendix A for the full concurrency matrix across HTTP, CLI, and MCP surfaces.

6.3 Port File

After binding, the server writes the actual port to `~/.punt-labs/quarry/data/<db>/serve.port`. This supports `--port 0` (ephemeral assignment). Removed on shutdown via lifespan context.

7 Configuration

7.1 Settings

All settings are environment variables read via pydantic-settings.

Variable	Default	Description
Paths		
QUARRY_ROOT	~/.punt-labs/quarry/data	Base directory for all databases
LANCEDB_PATH	<root>/default/lancedb	Vector database location
REGISTRY_PATH	<root>/default/registry.db	Directory sync SQLite database
LOG_PATH	~/.punt-labs/quarry/logs/quarry.log	Log file (rotating, 5 MB, 3 backups; independent of QUARRY_ROOT)
Chunking		
CHUNK_MAX_CHARS	1800	Target max characters per chunk (~450 tokens)
CHUNK_OVERLAP_CHARS	200	Overlap between consecutive chunks
Embedding		
EMBEDDING_MODEL	Snowflake/snowflake-arctic-embed-m-v1.5	Model identifier (display only — the ONNX model is fixed)
EMBEDDING_DIMENSION	768	Vector dimension (display only — fixed at 768)

7.2 Named Databases

`resolve_db_paths(settings, name)` maps a database name to a path under `QUARRY_ROOT`. Each database is fully isolated — its own LanceDB directory, sync registry, and vector index. Names containing path separators (`/` or `\`) or the literal traversal segments `.` or `..` are rejected; names like `foo.bar` are allowed. If `LANCEDB_PATH` was explicitly overridden, the override is preserved.

The persistent default database is stored in `~/.punt-labs/quarry/config.toml` as `[default] database = "name"`.

7.3 Chunking Tuning

Increase `CHUNK_MAX_CHARS` (3000–4000) for documents with long self-contained sections: legal contracts, academic papers, large source code functions.

Decrease `CHUNK_MAX_CHARS` (800–1200) for dense documents where each paragraph covers a distinct topic.

Increase `CHUNK_OVERLAP_CHARS` (400) when relevant information spans paragraph boundaries.

The embedding model has a 512-token context window. Chunks exceeding this are truncated during embedding, reducing quality for the tail.

8 Search and Retrieval

8.1 Embedding Model

snowflake-arctic-embed-m-v1.5 via ONNX Runtime:

- 768-dimensional vectors, 512-token context window
- Asymmetric retrieval: queries are prefixed with a search instruction, documents are not
- Output vectors are L2-normalized to unit length, so search runs under the **cosine** metric and similarity ($1 - \text{distance}$) is bounded true cosine in $[-1, 1]$ (DES-038)
- int8 quantized ONNX for CPU (~120 MB), FP16 for CUDA (~220 MB)
- Provider auto-detection: CUDA+FP16 when available, CPU+int8 fallback. `QUARRY_PROVIDER` env var overrides (`cpu`, `cuda`, or `unset` for auto). See DES-016 and `src/quarry/ingestion/provider.py`.
- Auto-downloads on first use via **huggingface.hub**

The model is fixed — there is no configuration to swap it. Changing the model invalidates all existing embeddings, requiring full re-ingestion.

8.2 OCR

RapidOCR handles all image-based text extraction. For semantic search, OCR accuracy matters less than expected — the embedding model is robust to minor OCR errors. A misspelled word rarely changes semantic meaning enough to affect search ranking.

8.3 Search Quality Tips

1. **Use natural language queries.** The model is trained on question-passage pairs. “What were Q3 revenue figures?” works better than “Q3 revenue.”
2. **Use collection filtering.** `quarry find "auth logic" -c backend`.
3. **Re-ingest after tuning.** Chunk parameters only affect new ingestions. Run `quarry sync` after changes.
4. **Check chunk boundaries.** `quarry find -n 1 "query"` and examine the result.
5. **Use `--overwrite` when re-ingesting.** Without it, duplicate chunks accumulate.

8.4 Hybrid Search

Hybrid search is the **default for every query** — there is no vector-only path. All three surfaces (CLI, HTTP, MCP) construct a `SearchService` (`src/quarry/retrieval/service.py`) and call its one `search` method, so the surfaces cannot drift into different retrieval behavior. This closes bug class 3 (remote/local divergence), where the HTTP `/search` endpoint once ran vector-only search while the CLI ran hybrid. `SearchService` wraps a `HybridRetriever` parameterized by a `RetrievalConfig` whose defaults reproduce the shipped pipeline bit-for-bit.

Channels:

1. **Vector similarity** — ANN search via LanceDB under the cosine metric
2. **BM25 full-text search** — Tantivy index on the `text` column, created during schema migration

Results are fused using Reciprocal Rank Fusion (RRF):

$$\text{score}(m) = \sum_{\text{channel}} \frac{1}{60 + \text{rank}_{\text{channel}}(m)}$$

FTS scoring. FTS rows arrive from Tantivy without a vector distance. Rather than displaying a synthetic 1.00 (which made irrelevant keyword hits appear as perfect matches), each FTS-only row is annotated with its *true* cosine distance against the query vector; a row whose stored vector is missing or zero-length is assigned the worst-case distance (similarity -1) instead.

Temporal decay is not a second retrieval channel — it is an additional weighting term applied only to agent-scoped memories (non-empty `agent_handle`). An `agent_handle` filter therefore narrows the result set and enables decay; it does not switch retrieval modes.

$$w = e^{-\lambda \cdot h}$$

where λ is the decay rate and h is hours since ingestion. Unscoped documents (`agent_handle = ""`) are exempt from decay (production leaves $\lambda = 0$).

8.5 Retrieval Seam and Evaluation

The `src/quarry/retrieval/` package is a deliberate seam (DES-037): the same `HybridRetriever` that serves production also serves the retrieval evaluation harness (`make eval`), so the committed baseline measures the retriever that actually ships. `RetrievalConfig` exposes the levers a bake-off varies — RRF k , over-fetch multiplier, distance metric, exact-vs-ANN vector search, reranker, and embedding strategy — without forking the code path. The harness is internal dev/CI tooling; it is not an end-user command. See `docs/eval-harness-design.md`.

Three columns extend the chunks table for agent-scoped memory:

Column	Type	Default	Purpose
<code>agent_handle</code>	utf8	""	Agent identity scope
<code>memory_type</code>	utf8	""	fact, observation, opinion, procedure
<code>summary</code>	utf8	""	One-line summary

Migration is idempotent — `_migrate_schema()` adds missing columns with empty-string defaults on every table open. Existing databases gain the columns transparently.

Identity tagging: the PreCompact hook reads the agent handle from `.punt-labs/ethos/config.yaml` and tags captured transcripts automatically.

8.6 Ethos Extension Integration

Quarry stores its configuration in ethos's generic extension mechanism (DES-008, DES-019):

```
~/.punt-labs/ethos/identities/<handle>.ext/quarry.yaml
```

The extension file contains two categories of data:

1. **Sidecar config** — `memory_collection` and `expertise_collections` are read directly by quarry via the filesystem (no ethos API call).

2. **Session context** — `session_context` is a YAML literal block scalar emitted verbatim by ethos at session start and before context compaction. It contains the agent’s memory instructions (collection name, slash commands, memory types).

`quarry install` (step 7/7) scans all identity extension directories and appends `session_context` to any `quarry.yaml` that has `memory_collection` but no `session_context`. Uses raw file append to preserve existing YAML comments and formatting. Per-identity exception handling ensures one malformed file does not abort the rest.

Ownership boundary: ethos emits quarry’s instructions without understanding them. Quarry writes its own instructions without importing ethos. The dependency is one-way — quarry depends on ethos for identity; ethos has zero knowledge of quarry’s internals.

9 Logging and Exceptions

9.1 Logging Standard

Every module that performs I/O declares `logger = logging.getLogger(__name__)`. Pure data modules (`models.py`, `types.py`, `chunker.py`) do not.

Logging is configured once at the application entry point (CLI or MCP server). Library modules never call `logging.basicConfig`.

Level	Use
DEBUG	Internal state: variable values, branch decisions, intermediate results.
INFO	Operational milestones: starting a job, completing a step, resource counts.
WARNING	Unexpected but recoverable: empty input, deprecated parameter usage.
ERROR	Failures that prevent completion but do not crash. Always with exception context.

CRITICAL is not used. If the process must exit, it raises an exception.

Log messages use `%s`-style formatting (lazy evaluation), not f-strings. Each message includes enough context to trace an operation without reading code.

9.2 Exception Handling

The codebase uses built-in exception types. No custom exceptions unless a caller needs to distinguish failure modes programmatically.

Exception	Raised when
<code>ValueError</code>	Invalid input: unsupported format, bad parameter.
<code>FileNotFoundError</code>	A file path argument does not exist.
<code>RuntimeError</code>	An operation reports failure (OCR engine error, ingestion failure).
<code>TimeoutError</code>	An external service exceeds its time budget.

Rules:

1. Never swallow exceptions. Every `except` block must re-raise, or log with `logger.exception()` and raise a different exception, or (at system boundaries only) return a documented sentinel.
2. Never use bare `except:`.
3. Never catch `Exception` broadly in library code. Broad catches are permitted only at the outermost boundary (MCP tool handler, CLI command).
4. Exceptions carry context: include the values that caused the failure.
5. Use `try/finally` for cleanup, not `try/except`.
6. Do not use exceptions for flow control.

10 Security

10.1 Bearer Token Authentication

`quarryd --api-key` (or `QUARRY_API_KEY` env var) gates all HTTP endpoints except `/health` behind `Authorization: Bearer <key>`.

- `/health` is exempt — load balancers need unauthenticated health checks.
- `OPTIONS` is exempt — CORS preflight requests carry no auth.
- `hmac.compare_digest` for token comparison — prevents timing attacks.
- Case-insensitive scheme per RFC 7235.
- Empty key = no auth — prevents accidental trivial bypass.
- No key = auth disabled — local use requires no key.

The server refuses to bind to a non-loopback address without `--api-key`.

10.2 WebSocket Security (CSWSH)

Origin-based cross-site WebSocket hijacking protection:

- If an `Origin` header is present, it must match the CORS allow-list.
- Browsers always send `Origin`; non-browser clients like `mcp-proxy` do not.
- Bearer auth is checked before WebSocket accept (close code 1008 on failure).

10.3 CORS

`CORSMiddleware` configured with GET and OPTIONS methods, Authorization and Content-Type headers. Default allowed origin: `http://localhost`. Configurable via `--cors-origin` (repeatable).

The server reflects the request's `Origin` header only when it matches the allow list, and adds `Vary: Origin` to signal cacheability varies by origin. When no `Origin` is present, no CORS headers are emitted.

10.4 TLS and TOFU Certificate Pinning

The installed service always runs with TLS enabled. `quarry serve` without `--tls` is supported for local development but must not be used for production. Certificate management uses TOFU (Trust On First Use) with self-signed CA pinning (DES-020).

Certificate generation: `quarry install` generates an EC P-256 CA and server certificate. Files are written atomically to `~/.punt-labs/quarry/tls/` with restricted permissions (0600 for keys, 0644 for CA cert). The server cert SAN includes the configured hostname (via `QUARRY_TLS_HOSTNAME`), `localhost`, and loopback addresses.

TOFU login:

1. Client fetches `/ca.crt` over HTTPS (verification disabled — bootstrap).
2. Displays SHA256 fingerprint for out-of-band confirmation.
3. Pins CA cert at `~/.punt-labs/mcp-proxy/quarry-ca.crt`.
4. All subsequent connections verify against the pinned CA only — system roots excluded.

mcp-proxy: The `quarry.toml` profile includes `ca_cert` pointing to the pinned CA. `mcp-proxy` builds a custom `tls.Config` with the pinned CA as the sole root, enforcing TLS 1.3 minimum.

10.5 Log Safety (CWE-532, CWE-117)

Uvicorn's access log is disabled (`access.log=False`), eliminating query string leakage at the source. The search handler logs only result count, never the raw query. WebSocket session keys are sanitized before logging: control characters stripped via `_CONTROL_CHAR_RE` and truncated to 64 characters to prevent log injection (CWE-117) and log flooding.

10.6 Capture PII Redaction (DES-036)

Captures — session transcripts and WebFetch auto-captures — are the input to the planned private shadow-repo sync (quarry-ow3k), so un-redacted content in a capture is a leak into a git-tracked (and soon pushed) surface. Every capture is scrubbed **at write time**, fail-closed, before any bytes reach disk.

Scrubber (src/quarry/scrub.py). A Scrubber applies redaction passes in a load-bearing order: *secrets* → *paths* → *emails* → *hostname* → *profanity*. Email must precede hostname so a hostname inside an email domain is subsumed by the whole-email redaction rather than half-redacted (which would leak the local part). Each pass emits a marker no pass can re-match, so scrubbing is idempotent — backfill may safely re-run.

- **Secrets** (src/quarry/scrub_rules.py): GitHub PATs, AWS keys, Anthropic/OpenAI keys, bearer headers, JWTs, PEM/GPG private key blocks, **KEY=value** env secrets, Slack tokens.
- **Paths**: home directories (/Users/<user>/, /home/<user>/) collapse to ~, retaining the useful deeper structure.
- **Emails**: any RFC-shaped address → [REDACTED:email]. A placeholder redacts every address (completeness beats attribution), not an ethos-handle mapping.
- **Hostname**: bounded to the local machine name (socket.gethostname() + .local + short leaf), not arbitrary dotted-token detection — the latter has a catastrophic false-positive rate on identifiers like github.com or quarry.db.facade.

Single write choke point (src/quarry/capture.py). Both .md producers — the PreCompact hook (src/quarry/hooks.py) and the backfill path (src/quarry/backfill.py) — route through one CaptureWriter. Scrubbing runs to completion before an atomic temp-file rename, so a scrub failure writes no file and a partial or half-redacted capture is never left behind.

WebFetch DB-ingest. The WebFetch auto-capture path scrubs page text via an opt-in content_scrubber parameter on ingest_url (src/quarry/ingestion/pipeline.py). The parameter defaults to None: user-initiated quarry ingest <url>, sitemap crawls, and directory sync are *not* scrubbed — redaction is a captures concern, not a general-ingestion one. The URL's persisted metadata form is redacted separately by CaptureUrl (src/quarry/capture_url.py), which strips userinfo, query, and fragment (preserving IPv6 brackets) before the same text scrubber runs over the bare scheme://host/path.

11 Deployment

11.1 Single Machine

```
curl -fsSL https://raw.githubusercontent.com/.../install.sh | sh
```

Installs quarry, embedding model, mcp-proxy, Claude Code plugin, and registers a daemon with TLS on localhost.

11.2 Network Mode (Remote GPU)

```
export QUARRY_API_KEY=$(openssl rand -hex 32)
curl -fsSL https://raw.githubusercontent.com/.../install.sh | sh -s -- --network
```

Same as default, but binds daemon to 0.0.0.0:8420 instead of localhost. Requires QUARRY_API_KEY. Installs quarry, embedding model, generates TLS certificates, registers systemd service. Detects NVIDIA GPUs and swaps onnxruntime for onnxruntime-gpu. Prints CA fingerprint for client TOFU. Claude Code plugin installed if claude CLI is on PATH.

11.3 Client (Remote Access)

```
curl -fsSL https://raw.githubusercontent.com/.../install.sh | sh
quarry login <server> --api-key <token>
```

No special flag needed. Default install runs a local daemon on localhost. quarry login redirects queries to the remote server via TOFU certificate pinning and writes the mcp-proxy config.

11.4 Daemon (quarryd)

```
QUARRY_API_KEY=<token> quarryd --port 8420 --host 0.0.0.0 --tls
```

Non-loopback hosts require an API key (the server refuses to start without one).

Managed by `launchd` (macOS) or `systemd` (Linux) via `quarry install`. API key delivered via `systemd EnvironmentFile` (Linux) or `launchd EnvironmentVariables` (macOS) — never in process arguments.

11.5 Container (Fly.io)

Multi-stage Dockerfile, three stages:

1. **deps** (`python:3.13-slim`): installs `uv`, runs `uv sync --frozen --no-dev`. Two-pass for Docker layer caching (deps first, then source).
2. **model** (extends **deps**): downloads the embedding model at build time (~120 MB, int8 quantized). Cached as a separate Docker layer.
3. **runtime** (`python:3.13-slim`): copies `/app` from **deps**, copies model cache from **model** stage. `QUARRY_ROOT=/data`, port 8080.

LanceDB data lives on a Fly persistent volume at `/data`. `QUARRY_API_KEY` is set via `fly secrets`. Cold start is ~5s with the baked-in model.

12 Performance

12.1 Critical Path

Operation	Latency	Notes
Model load	~250 ms	One-time per process
Embedding (1 chunk)	~50 ms	Apple Silicon, CPU
LanceDB search	<10 ms	Vector similarity + filter
End-to-end search	<100 ms	After model is loaded

12.2 Fire-and-Forget MCP Pattern

Side-effect MCP tools return immediately and process in a bounded thread pool (4 workers). This is necessary because:

1. MCP tool calls block the LLM's response stream. A 30-second ingest blocks Claude from responding.
2. The bounded pool prevents resource exhaustion under concurrent requests.
3. Settings and database connections are snapshotted at the tool boundary, avoiding race conditions with the mutable `_db_name` `ContextVar`.

12.3 Single Table Design

All chunks live in one LanceDB table (`chunks`). Document and collection boundaries are columns, not separate tables. This simplifies cross-document search and avoids table proliferation. Filtering by `document/collection/page_type/source_format` uses LanceDB's built-in filter predicates on the vector search.

13 Plugin and Hook System

The Claude Code plugin (`.claude-plugin/`) registers hooks that fire on session events and tool calls. All hooks are fail-open: exceptions are logged and the process exits 0.

Hook Event	Matcher	Effect	Blocks?
SessionStart	*	Auto-register directory + background sync	No
PostToolUse	quarry tools	Suppress tool output formatting	No
PostToolUse	WebFetch	Auto-ingest fetched URL (scrubbed)	No
PreCompact	*	Capture transcript as session notes (scrubbed)	No

13.1 Session Start

`handle_session_start()` auto-registers the working directory and launches a background sync via a detached subprocess. Uses atomic PID-file locking (`O_CREAT|O_EXCL`) at `~/.punt-labs/quarry/sync.pid` to prevent concurrent syncs. Returns `additionalContext` immediately without waiting.

13.2 Web Fetch Capture

`handle_post_web_fetch()` ingests URLs fetched during a session into the `web-captures` collection. Deduplicates by document name. Prefers already-fetched content from the tool response (avoids re-fetch, reduces SSRF surface). Both ingress branches (primary and the `ingest_url` re-fetch fallback) scrub page text and redact the URL metadata before content reaches the pushable `web-captures` collection (DES-036).

13.3 Pre-Compact Capture

`handle_pre_compact()` reads the transcript (JSONL) and ingests it into the `session-notes` collection. Extracts user/assistant message text, skipping tool-use blocks. Capped at 500,000 characters. The capture file is written through the shared `CaptureWriter`, which scrubs secrets, PII, and profanity before the atomic write (DES-036).

14 CLI Commands

Command	Purpose
<code>ingest <source></code>	Ingest file or URL. Flags: <code>--agent-handle</code> , <code>--memory-type</code> , <code>--summary</code>
<code>remember</code>	Ingest stdin content
<code>find <query></code>	Semantic search
<code>show <document></code>	Document metadata or page text
<code>status</code>	Database statistics
<code>use <name></code>	Set persistent default database
<code>delete <name></code>	Delete document or collection
<code>register <directory></code>	Register for incremental sync
<code>deregister <collection></code>	Remove directory registration
<code>sync</code>	Sync all registered directories
<code>enable [directory]</code>	Enable quarry knowledge capture for a project directory
<code>disable [directory]</code>	Disable quarry knowledge capture; remove registration and data
<code>optimize</code>	Compact LanceDB table and rebuild indexes (CLI-only). Flags: <code>--force</code>
<code>list documents</code>	List indexed documents

Command	Purpose
<code>list collections</code>	List collections with counts
<code>list databases</code>	List named databases
<code>list registrations</code>	List registered directories
<code>serve</code>	Start HTTP API server
<code>mcp</code>	Start MCP server (stdio)
<code>login <host></code>	TOFU login to remote server (cert pinning)
<code>logout</code>	Disconnect from remote server
<code>remote list</code>	Show remote config; <code>--ping</code> validates connection
<code>install</code>	Set up data directory, download model, generate TLS certs, register daemon, detect GPU, configure ethos ext
<code>doctor</code>	Check environment health
<code>uninstall</code>	Remove system daemon
<code>version</code>	Print version

Global flags: `--json`, `--verbose/-v`, `--quiet/-q`, `--db <name>`.

Remote routing (DES-021): When a remote server is configured (via `quarry login`), all data commands should route to the remote server over HTTPS with the pinned CA cert. Only authentication commands (`login`, `logout`, `remote`) and administration commands (`install`, `uninstall`, `serve`, `mcp`, `version`) are local-only. All data commands — `find`, `show`, `status`, `ingest`, `remember`, `delete`, `sync`, every `list` subcommand, and directory `register/deregister` — route remotely. `use` is local-only by design (the remote server is fixed to its own database).

15 Test Architecture

Roughly 1,500 tests across the suite. Three tiers plus integration:

Tier	Strategy	Files
Tier 1: Core logic	Real data, no mocks	test_database, test_chunker, test_embeddings, test_code_processor, test_html_processor, test_text_processor, test_collections, test_config, test_formatting, test_image_analyzer, test_latex_utils, test_models, test_pdf_analyzer, test_presentation_processor, test_spreadsheet_processor, test_text_extractor, test_registry (17 files)
Tier 2: Orchestration	Mocked I/O	test_pipeline, test_pipeline_images, test_ocr_local, test_url_ingestion, test_sitemap, test_backends, test_sync, test_hooks, test_doctor (9 files)
Tier 3: Surface wiring	Fully mocked, <2s	test_cli, test_mcp_server, test_mcp_websocket, test_http_server, test_proxy, test_service (6 files)
Integration	Real ONNX + real LanceDB	test_integration (@pytest.mark.slow)

Quality gates: **ruff** (lint + format), **mypy** (strict), **pyright** (strict), **pytest** (all markers except slow/integration). Zero violations required.

16 Formal Model

The plugin state machine — how quarry hooks into the Claude Code session lifecycle — is formally specified in `docs/claude-code-quarry.tex` using Z notation (ISO 13568).

The specification covers:

- Session start with/without auto-sync
- Web fetch auto-capture into **web-captures** collection
- Pre-compaction transcript capture into **session-notes**
- Database switching via **use** tool
- Knowledge hint injection at tool boundaries
- Compaction observability: a capture flag records whether capture preceded compaction; the hook system enforces capture-before-compaction on a best-effort basis
- Clean session boundaries: all quarry state cleared on **EndSession**

The specification can be type-checked with `fuzz -t claude-code-quarry.tex`.

A Operation Concurrency Model

Every quarry operation is exposed on up to three surfaces: the HTTP API (daemon), the CLI, and the MCP server (stdio or WebSocket). The concurrency model differs by surface because each has different constraints: HTTP connections time out, MCP tool calls block the LLM response stream, and CLI commands run in a user’s terminal.

Table 4 classifies each operation. The **Client model** column describes what the caller experiences:

sync	The caller blocks until the operation completes and receives the result inline.
fire-and-forget	The caller receives an acknowledgement immediately (HTTP 202 or a short string). Work runs in the background; the caller polls or ignores the outcome.
—	The operation is not available on that surface.

A.1 Concurrency Matrix

Table 4: Operation concurrency by surface. *Long-running* = involves network I/O, embedding, or bulk LanceDB writes.

Operation	HTTP API	CLI (local)	CLI (remote)	MCP
<i>Read operations</i>				
find (search)	sync	sync	sync	sync
show	sync	sync	sync	sync
status	sync	sync	sync	sync
list documents	sync	sync	sync	sync
list collections	sync	sync	sync	sync
list registrations	sync	sync	sync	sync
list databases	sync	sync	sync	sync
health	sync	—	—	—
<i>Mutating operations</i>				
register	fire-and-forget	sync	fire-and-forget	fire-and-forget
deregister	fire-and-forget	sync	fire-and-forget	fire-and-forget
delete (document)	fire-and-forget	sync	fire-and-forget	fire-and-forget
delete (collection)	fire-and-forget	sync	fire-and-forget	fire-and-forget
remember	fire-and-forget	sync	fire-and-forget	fire-and-forget
ingest (URL)	fire-and-forget	sync	fire-and-forget	fire-and-forget
ingest (file)	—	sync	—	fire-and-forget
sync	fire-and-forget	sync	fire-and-forget	fire-and-forget
use (switch db)	sync	sync	sync	sync
optimize	—	sync	—	—

A.2 Design Rationale

Read operations are synchronous on all surfaces. Hybrid search (embedding + vector similarity + BM25) completes in <100 ms after model load; the overhead of task tracking would exceed the operation itself.

Mutating operations are all fire-and-forget on HTTP. Every mutating endpoint returns 202 Accepted with a **task_id**; work runs in a background **asyncio.Task**. This eliminates the 15-second CLI timeout as a failure mode for any write operation.

- All mutating endpoints use **GET /tasks/{task_id}** for status polling. **/sync/{id}** and **/ingest/{id}** are backwards-compatible aliases.
- **Sync** rejects concurrent requests with HTTP 409 because a second sync against the same registrations would race on LanceDB writes. All other operations allow concurrency.

- Completed and failed tasks are garbage-collected after a 1-hour TTL on the next task creation.
- **Optimize** is CLI-only (no HTTP or MCP surface). It runs LanceDB compaction and index rebuilds, which are CPU-intensive and should not be triggered remotely without explicit intent.

A.3 Task Lifecycle

All background tasks follow the same lifecycle:

1. `POST` validates input, creates a `TaskState` record on the server context, spawns an `asyncio.Task` wrapping `run_in_threadpool()`, and returns HTTP 202 with `task_id` and `status = accepted`.
2. The background task updates `status` to `completed` (with `results`) or `failed` (with `error`) on completion.
3. A `finally` guard marks any task that exits without setting a terminal status as `failed`.
4. `CancelledError` is caught before `Exception` to ensure clean shutdown on server stop.
5. The CLI prints the `task_id` and exits 0 (fire-and-forget).